CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Earthquake Platform

Goal: Build and improve a model structure that remains stable during a repeatable shake-table test.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 10, on-track choice

Ask | Plan | Build | Test | Improve

You are done when

Compare the two damage scores and name the structural change supported by evidence.

You will use

- Shake platforms
- Craft sticks
- Straws
- Cards
- Tape
- Binder clips
- Damage rubric

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the platform on its boundary mat.
- Check the height rule.

2 Step 2: Platform safety — 3 min

- Keep fingers outside the moving platform.
- Test only on the teacher signal.
- Stop any falling part.

3 Step 3: Plan for shaking — 4 min

- Sketch the structure.
- Mark two likely weak joints.
- Check the required footprint.

4 Step 4: Build the structure — 9 min

- Build the base inside the footprint.
- Reach the required height.
- Check every connection.

5 Step 5: Run the shake — 7 min

- Move hands behind the boundary.
- Run the standard shake sequence.
- Record movement or damage.

6 Step 6: Strengthen one part — 4 min

- Improve one weak connection.
- Repeat the same shake sequence.

7 Step 7: Reset the platform — 2 min

- Remove loose pieces from the platform.
- Sort reusable parts.

8 Step 8: Exit check — 3 min

- Compare both damage scores.
- Name the stronger feature.

What to turn in

Two damage scores and the change that helped.

No computer? Another way to do it

Shake the table by hand with a fixed count.

Finished early? Next challenge

Aftershock: Test whether the improved structure survives a second shake without repairs.

Stay safe

Keep fingers outside the moving platform boundary and test only on the teacher signal.

Grade targets

Grade 5: Build, shake, fix one weak spot.
Grade 6: Compare damage scores and name the structural change.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.